An Overview of the Service Discovery Feature in Hyperledger Fabric

This article will appeal to open source technology enthusiasts interested in blockchain technologies and with a working knowledge of the Hyperledger Fabric project.

Prior to the release of Hyperledger Fabric v1.2, the SDK needed a lot of information in order to allow applications to connect to the relevant network nodes. This information was statically encoded. However, this implementation was not dynamically reactive to network changes like the addition of peers who have installed the relevant chaincode, or are temporarily offline. Static configurations also do not allow applications to react to changes in the endorsement policies itself, which might happen when a new organisation joins a channel. The client application has no way of knowing which peers have updated ledgers and which do not. The application might submit proposals to peers whose ledger data is not in sync with the rest of the network, resulting in transactions being invalidated upon commit and resources getting wasted as a consequence. In order to execute chaincode on peers, we need to submit transactions to orderers, and these need to be updated about the status of transactions as well as the applications connect to an API exposed by an SDK.

The discovery service improves this process by having the peers compute the needed information dynamically and presenting it to the SDK in a consumable manner.

Service discovery in Fabric

The application issues a configuration query to the discovery service and obtains all the static information it would have otherwise needed to communicate with the rest of the nodes of the network. This information can be refreshed at any time by sending a subsequent query to the discovery service of a peer. The application is bootstrapped, knowing about a group of peers which is trusted by the application developer/administrator to provide authentic responses to discovery queries. A good candidate peer to be used by the client application is one that is in the same organisation.